Progressive Education Society's
Modern College of Engineering, Pune-05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Department of S.Y. B. Tech [AIDS]

Assignment No: 11

SUBJECT: Java Programming

Name: Shravan Kalbhor
Class: SY BTech (AIDS)
Section: A
Roll Number: 26026

Title / Problem Statement :

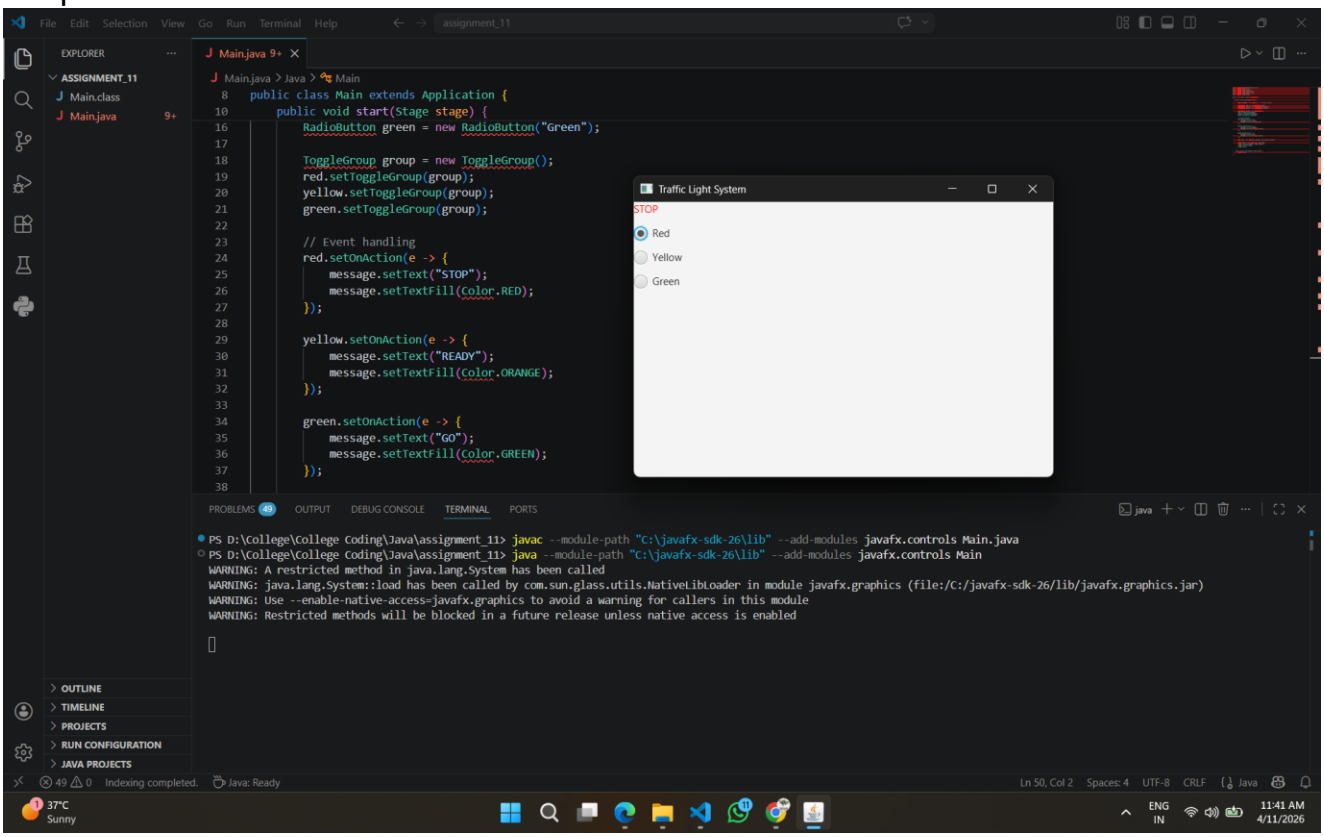
Develop JavaFX application that simulates a traffic light system. The GUI lets the user select one of three lights: red, yellow or green with radio buttons. on selecting a button an appropriate message with "stop" or "ready" or "go" should appear above the buttons in a selected color, initially there is no message shown.

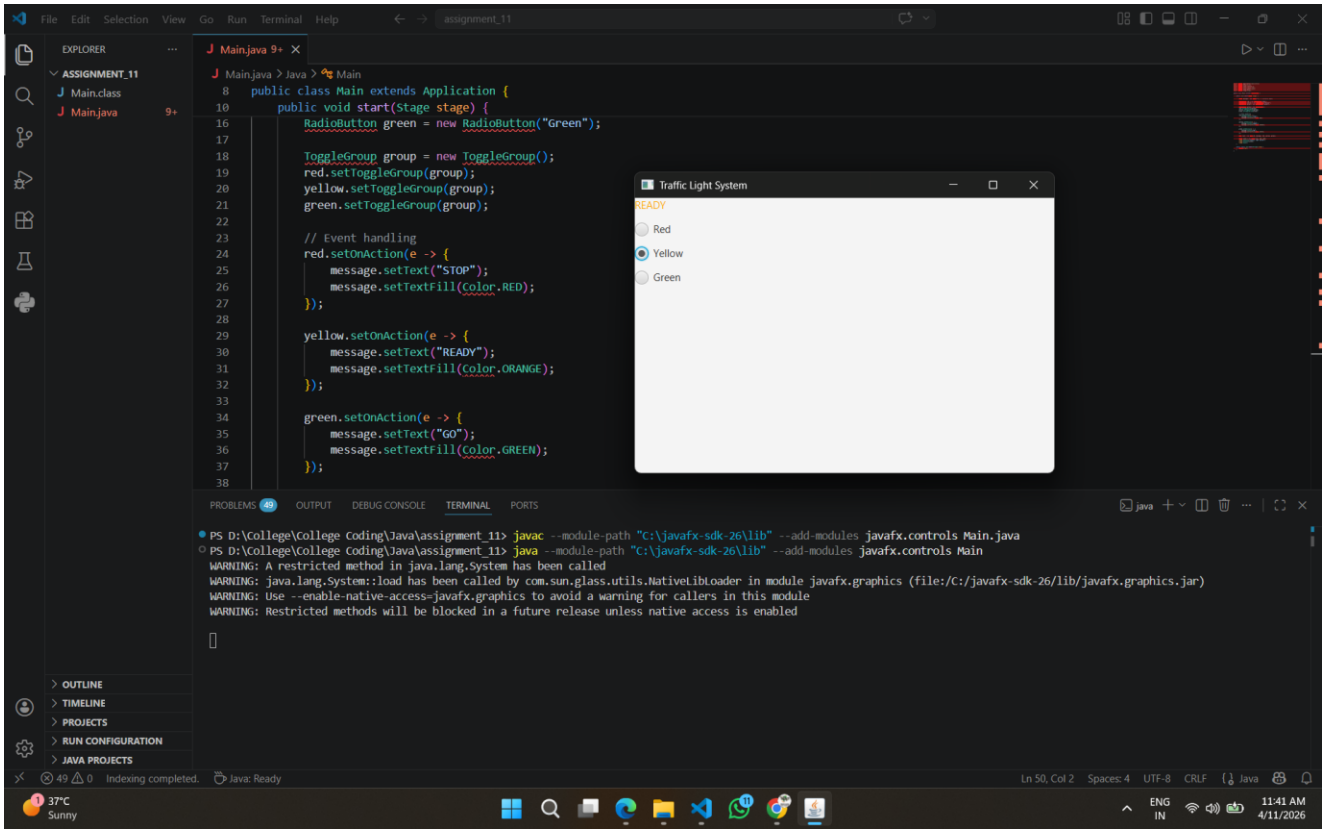
Input :

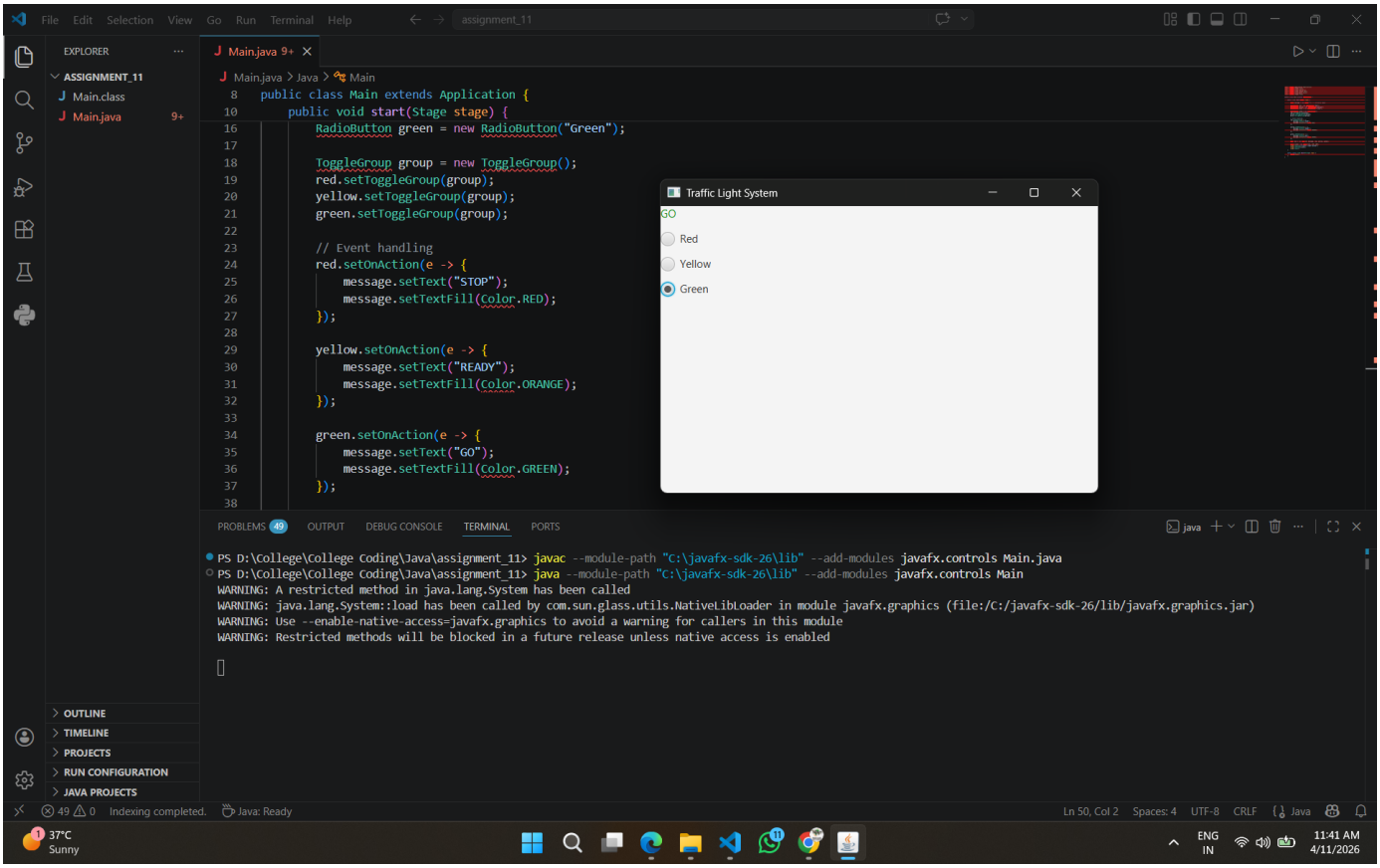
```java
import javafx.application.Application;

import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.VBox;
import javafx.scene.paint.Color;
import javafx.stage.Stage;

public class Main extends Application {

    public void start(Stage stage) {

        Label message = new Label(""); // initially empty

        RadioButton red = new RadioButton("Red");
        RadioButton yellow = new RadioButton("Yellow");
        RadioButton green = new RadioButton("Green");

        ToggleGroup group = new ToggleGroup();
        red.setToggleGroup(group);
        yellow.setToggleGroup(group);
        green.setToggleGroup(group);

        // Event handling
        red.setOnAction(e -> {
            message.setText("STOP");
            message.setTextFill(Color.RED);
        });

        yellow.setOnAction(e -> {
            message.setText("READY");
            message.setTextFill(Color.ORANGE);
        });

        green.setOnAction(e -> {
            message.setText("GO");
            message.setTextFill(Color.GREEN);
        });
```

```
        VBox root = new VBox(10, message, red, yellow, green);

        Scene scene = new Scene(root, 250, 200);
        stage.setTitle("Traffic Light System");
        stage.setScene(scene);
        stage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Output :


Main.java 9+
Main.java > Java > Main
public class Main extends Application {
public void start(Stage stage) {
RadioButton green = new RadioButton("Green");
ToggleGroup group = new ToggleGroup();
red.setToggleGroup(group);
yellow.setToggleGroup(group);
green.setToggleGroup(group);
// Event handling
red.setOnAction(e -> {
message.setText("STOP");
message.setTextFill(Color.RED);
});
yellow.setOnAction(e -> {
message.setText("READY");
message.setTextFill(Color.ORANGE);
});
green.setOnAction(e -> {
message.setText("GO");
message.setTextFill(Color.GREEN);
});
Traffic Light System
STOP
Red
Yellow
Green
PS D:\College\College Coding\Java\assignment_11> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_11> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled


Main.java 9+
Main.java > Java > Main
public class Main extends Application {
public void start(Stage stage) {
RadioButton green = new RadioButton("Green");
ToggleGroup group = new ToggleGroup();
red.setToggleGroup(group);
yellow.setToggleGroup(group);
green.setToggleGroup(group);
// Event handling
red.setOnAction(e -> {
message.setText("STOP");
message.setTextFill(Color.RED);
});
yellow.setOnAction(e -> {
message.setText("READY");
message.setTextFill(Color.ORANGE);
});
green.setOnAction(e -> {
message.setText("GO");
message.setTextFill(Color.GREEN);
});
Traffic Light System
READY
Red
Yellow
Green
PS D:\College\College Coding\Java\assignment_11> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_11> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled


File Edit Selection View Go Run Terminal Help
assignment_11
EXPLORER
ASSIGNMENT_11
Main.class
Main.java 9+
Main.java > Java > Main
public class Main extends Application {
public void start(Stage stage) {
RadioButton green = new RadioButton("Green");
ToggleGroup group = new ToggleGroup();
red.setToggleGroup(group);
yellow.setToggleGroup(group);
green.setToggleGroup(group);
// Event handling
red.setOnAction(e -> {
message.setText("STOP");
message.setTextFill(Color.RED);
});
yellow.setOnAction(e -> {
message.setText("READY");
message.setTextFill(Color.ORANGE);
});
green.setOnAction(e -> {
message.setText("GO");
message.setTextFill(Color.GREEN);
});
Traffic Light System
GO
Red
Yellow
Green
PROBLEMS 49
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS D:\College\College Coding\Java\assignment_11> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_11> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
Indexing completed.
Java: Ready
Ln 50, Col 2
Spaces: 4
UTF-8
CRLF
Java
37°C
Sunny
ENG
IN
11:41 AM
4/11/2026